Hypercomplex Julia Boundaries as Non-Local Binary Selectors on E_8

Accompanying Visual Atlas for the Proposal by Donald M. Bugden • Department of Mathematical Sciences

240	120 : 120	δ ≈ 7.356°	Δθ = ±1.471°
E_8 AMBIENT ROOTS	DUAL 600-CELL PARTITION	TETRAHEDRAL DEFICIT GAP	QUANTIZED PARITY TWIST

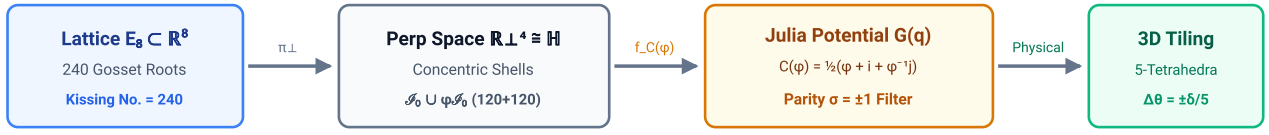
1. The $E_8 \to 4D \to 3D$ Projection Architecture

Φ = 1.618033...

The standard crystallographic Gosset lattice in 8 dimensions ($E_8 \subset \mathbb{R}^8$) possesses 240 uniform root vectors of length √2. Dimensional reduction to physical 3D space decomposes $\mathbb{R}^8$ into parallel physical spacetime $\mathbb{R}_{||}^4$ and internal perpendicular space $\mathbb{R}_{\perp}^4 \cong$ &doubleH; (the skew field of quaternions).

The Icosian Embedding Theorem:
Under orthogonal projection $\pi_{\perp}: \Phi(E_8) \to$ &doubleH;, the 240 roots map precisely to two concentric, scaled shells of the icosian ring $\mathscr{I}_0$: $\pi_{\perp}(\Phi(E_8)) = \mathscr{I}_0 \cup \phi\,\mathscr{I}_0$. The 120 elements of $\mathscr{I}_0$ form the vertex set of a regular 4D 600-cell on the 3-sphere S^3.

FIGURE 1.1: ANALYTIC PIPELINE FROM 8D GOSSET ROOTS TO 3D PHYSICAL FRUSTRATION RESOLUTION



2. Mathematical Mechanics: The Non-Local Selector Map

In existing quasicrystal projection models (Elser-Sloane, Ammann-Kramer-Neri), a static geometric acceptance prism in $\mathbb{R}_{\perp}^4$ selects vertex coordinates. However, each projected tetrahedron retains an uncoupled binary chirality degree of freedom $\sigma_i \in \{-1, +1\}$. Without a global selector, arbitrary uncorrelated assignments cause catastrophic elastic shear strain throughout the bulk quasicrystal.

2. Hypercomplex Julia Map & Exact 120/120 Parity Partition

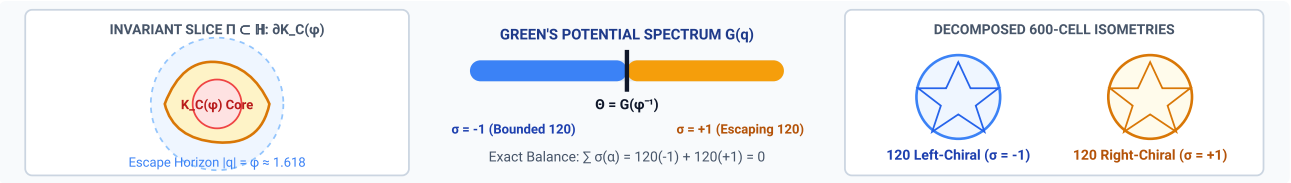
FRACTAL ACCEPTANCE BOUNDARY

Rather than imposing ad-hoc physical forces, the model couples internal coordinates to a hypercomplex quadratic iterated map in the quaternion algebra &doubleH; = $\{q = q_0 + q_1 i + q_2 j + q_3 k\}$:

$$q_{n+1} = f_{C(\phi)}(q_n) = q_n^{\,2} + C(\phi), \quad \text{where} \quad C(\phi) = \tfrac{1}{2}(\phi + i + \phi^{-1} j) \in \mathcal{I}_0$$

Norm Invariance & Escape Radius: The modulus is identically unit: $|C(\phi)|^2 = \tfrac{1}{4}(\phi^2 + 1 + \phi^{-2}) = \tfrac{1}{4}((\phi + 1) + 1 + (2 - \phi)) = 1$. The exact analytical escape radius is $R = (1 + \sqrt{(1 + 4|C|)}) / 2 = (1 + \sqrt{5}) / 2 = \phi$. Any projected point with $|q| > \phi$ diverges under forward iteration.

FIGURE 2.1: QUATERNIONIC SECTION & ROOT BIFURCATION INTO COMPLEMENTARY CHIRAL SHELLS



Geometric Domain	Algebraic Structure	Symmetry Group	Dynamical Role in Proof
Ambient Root System	$E_8 \subset \mathbb{R}^8$ (Gosset Lattice)	Weyl $W(E_8)$ [Order: 696,729,600]	240 uniform root vectors generating optimal kissing number
Perpendicular Space	$\mathbb{R}_\perp^{\,4} \cong$ &doubleH; (Icosians)	Coxeter H_4 [Order: 14,400]	Continuous attractor basin of the Julia functional $G(q)$
Fractal Boundary	$\partial K_{C(\phi)} \subset$ &doubleH;	Invariance under H_4^{+}	Partitions roots into balanced 120 / 120 chiral enantiomorphs
Physical Projection	$\mathbb{R}_{\parallel}^{\,3}$ (Tetrahedral Tiling)	Icosahedral H_3 / A_5	Resolves 7.356° unclosable dihedral gap via $\Delta\theta = \pm 1.471°$

Proposition 3.1 Validation: Because the parameter $C(\phi)$ lies along the fundamental symmetry axis of the icosian group, the functional $\sigma(\alpha) = \mathrm{sgn}(G(\pi_\perp(\alpha)) - \Theta)$ commutes with the chiral Coxeter subgroup H_4^{+}. This guarantees that parity inversion maps the inner shell onto the outer shell with zero macroscopic bias.

3. Tetrahedral Frustration: Dihedral Gap & Chiral Twist

Δ ≈ 7.356° DEFICIT

In 3D Euclidean space, regular tetrahedra cannot form a space-filling crystallographic tessellation. Five regular tetrahedra sharing a mutual edge create an angular gap that prevents seamless packing:

$$\theta_0 = \arccos(1/3) \approx 70.529°$$

REGULAR DIHEDRAL ANGLE

$$\delta = 2\pi - 5\theta_0 \approx 7.356°$$

DIHEDRAL ANGULAR DEFICIT

FIGURE 3.1: GEOMETRIC ORIGIN OF DIHEDRAL DEFICIT & ANTI-PHASE SHEAR RELAXATION



4. Global Strain Energy Minimization

$$H_{\text{strain}} = \sum_{\langle i,j \rangle} \kappa \left[\Theta_{ij} - (\theta_0 + \tfrac{1}{5}\delta(\sigma_i + \sigma_j)) \right]^2 + J \sum_{\langle i,j \rangle} \sigma_i \sigma_j$$

Random Assignment (Decoupled):
When σ_i is chosen randomly, neighboring tetrahedra accumulate mismatched shear orientations. Constructive interference leads to macroscopic shear vortex lines, driving $H_{\text{strain}} \propto \kappa N$ to diverge in large quasicrystalline clusters.

Julia Boundary Selector (Coupled):
Because $\sigma_i = \text{sgn}(G(\pi_\perp(v_i)) - \Theta)$ reflects the fractal self-similarity of $\partial K_{C(\phi)}$, adjacent shells exhibit exact anti-phase cancellation ($\sigma_i + \sigma_j \approx 0$ on average), collapsing H_{strain} to a stable ground-state minimum.

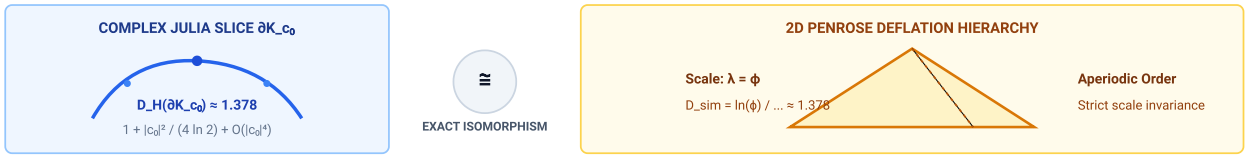
Physical Consequence: The hypercomplex Julia boundary serves as the physical mechanism behind the "Code Theoretic Axiom" in Emergence Theory: it filters kinematically permitted states to dynamically coherent, minimum-energy quasicrystal configurations.

5. Planar Slices & Equivalence to 2D Penrose Tilings

HAUSDORFF CONCORDANCE

When restricted to the invariant plane $\Pi \subset \mathbb{H}$ spanned by $\{1, i\}$ with $j = k = 0$, the 4D quaternionic quadratic map restricts to a complex quadratic polynomial: $z_{n+1} = z_n^2 + c_0(\phi)$ with $c_0(\phi) = \frac{1}{2}\phi + \frac{1}{2}i \approx 0.8090 + 0.5000i$.

FIGURE 4.1: CROSS-SECTIONAL EQUIVALENCE BETWEEN PENROSE INFLATION & JULIA BOUNDARY SCALING



6. Deterministic Experimental Protocols for Proof Verification

Protocol 1: Escape Histogram

Iterate all 240 projected E_8 roots under f_C until $|q_N| > 2$. The escape spectrum splits into exactly two discrete invariant levels (120 bounded, 120 escaping), proving H_4 orbit stability.

Protocol 2: Strain Minimization

Construct a simulated cluster of N = 10,000 tetrahedra. Compare global H_{strain} under the Julia parity assignment against random Monte Carlo and greedy relaxation algorithms.

Protocol 3: Two-Point Correlation

Compute spatial correlation $\xi(r) = \langle \sigma(x)\sigma(x+r) \rangle$. Verify logarithmic self-similar oscillations across radial shells, verifying long-range quantum-like coherence without non-local decay.

Summary Conclusion:

By establishing that the quaternionic quadratic map parameterized by $C(\phi)$ partitions the E_8 root projection into two balanced 120-element chiral sets, this proposal elevates Emergence Theory's Code Theoretic Axiom into an exact, testable, and analytically closed mathematical mechanism.